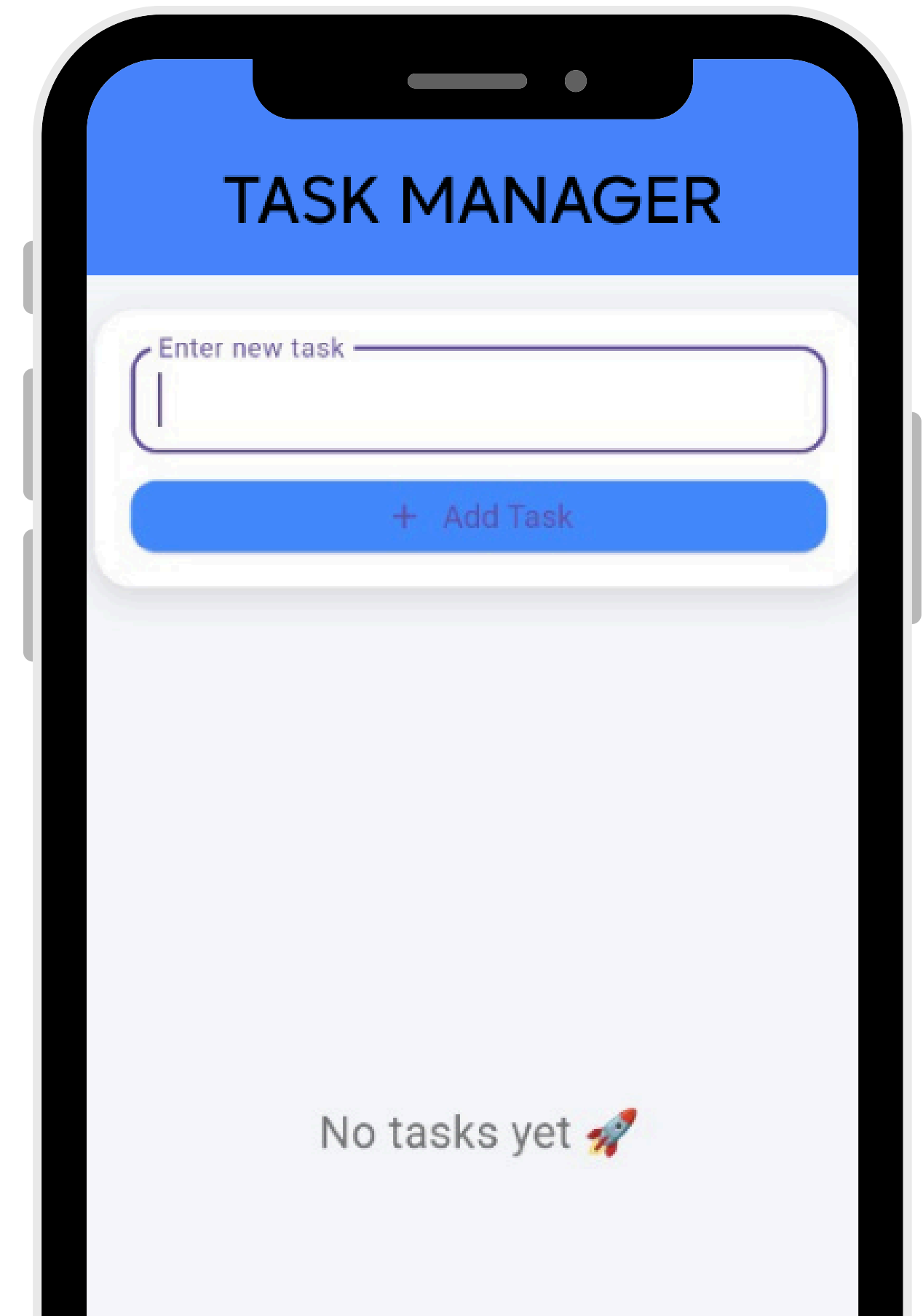


Dart and Flutter

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Course: CPL



Project Goals

01

Brief Explanation
of the App

02

key programming
constructs

01

Dart Console
Version

02

Flutter Mobile
Version

01

Architecture

02

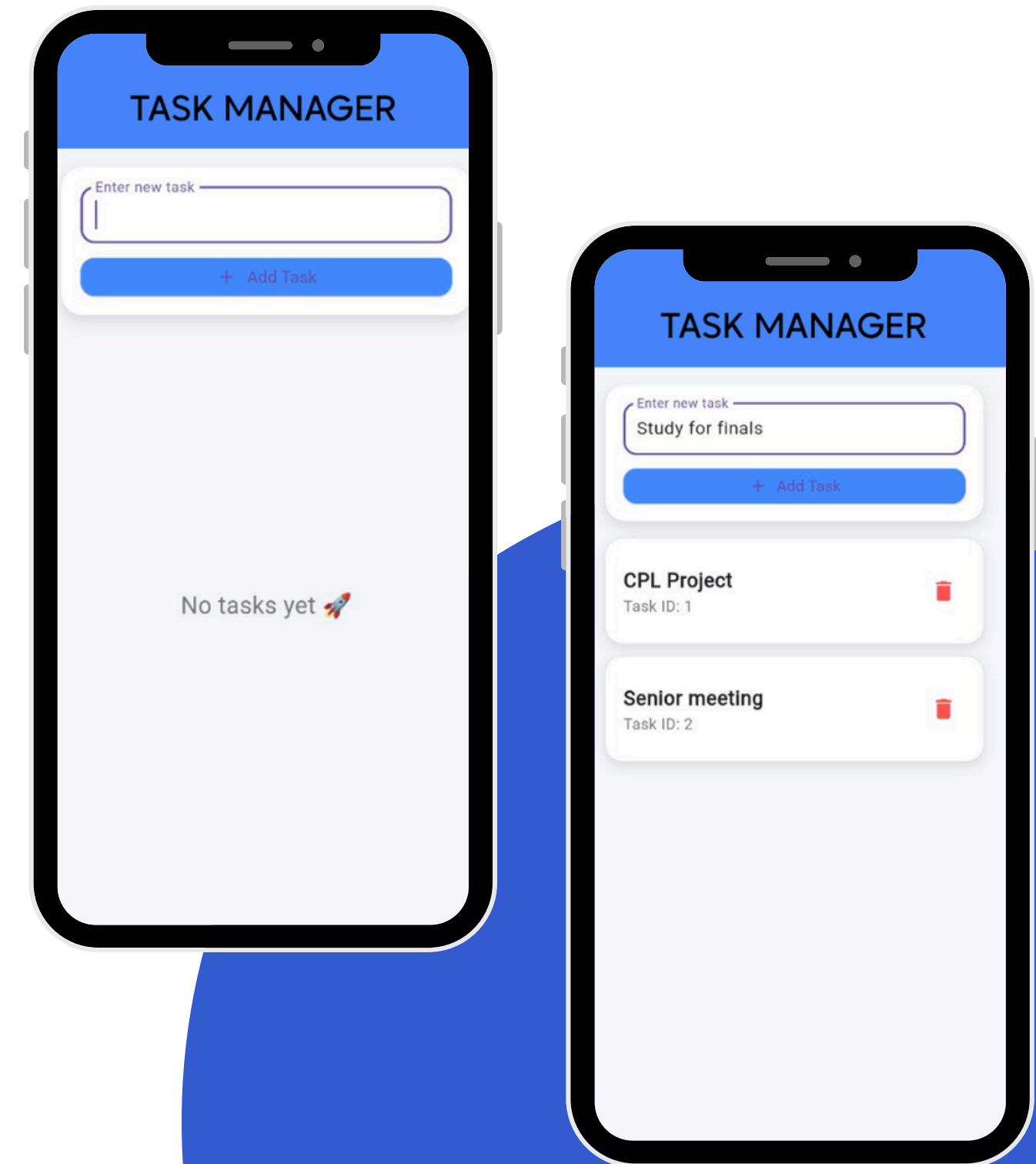
Language
Comparison

01

Final App Demo

02

Challenges &
Solutions



Application Service

Project Goals

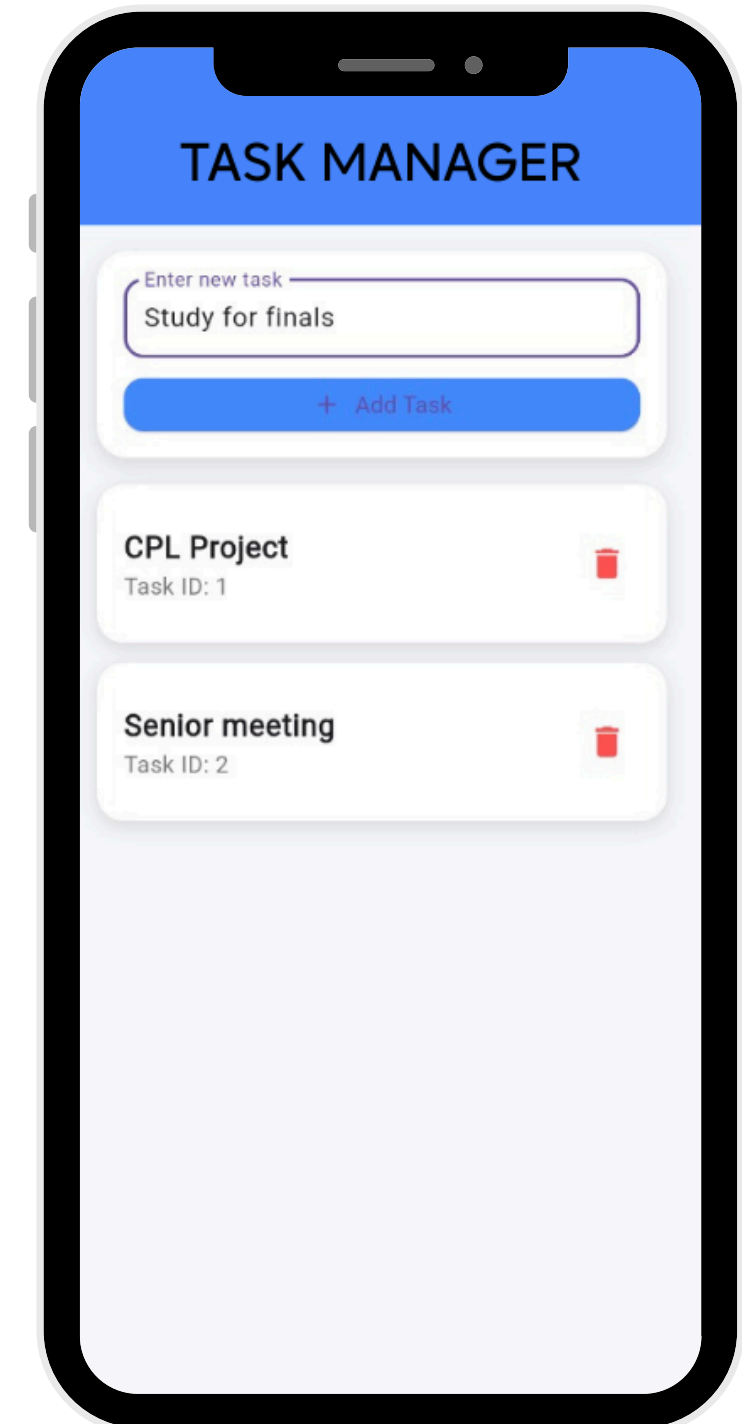
Our project aims to:

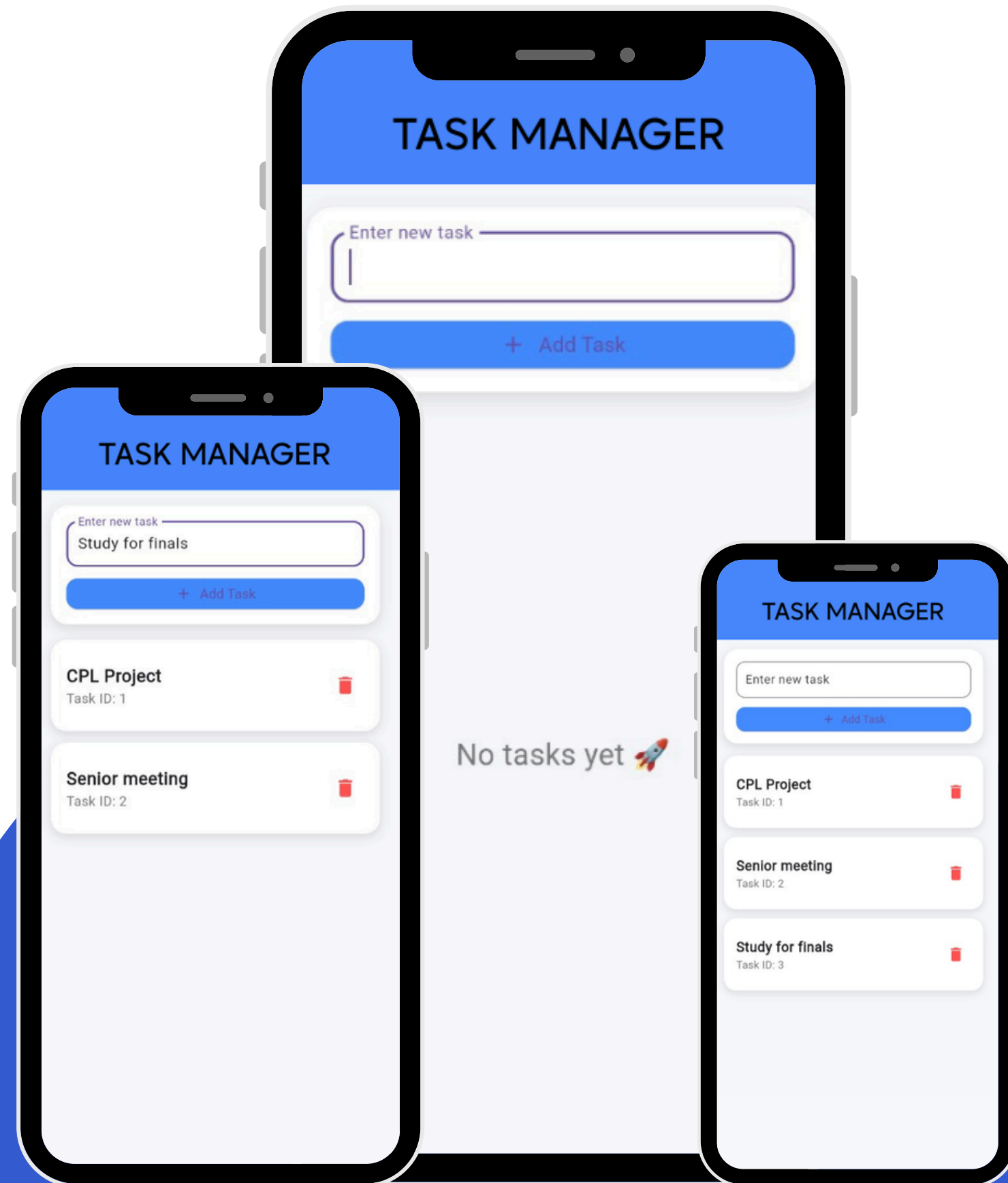
- Demonstrate programming constructs
- Apply them in a real-world application
- Compare Dart with Java
- Build two implementations:
 - Dart console version
 - Flutter mobile app version

Why We Chose Dart

Dart is:

- Easy to learn (similar to Java)
- Modern and clean syntax
- Great for cross-platform development
- Works perfectly with Flutter to create mobile apps



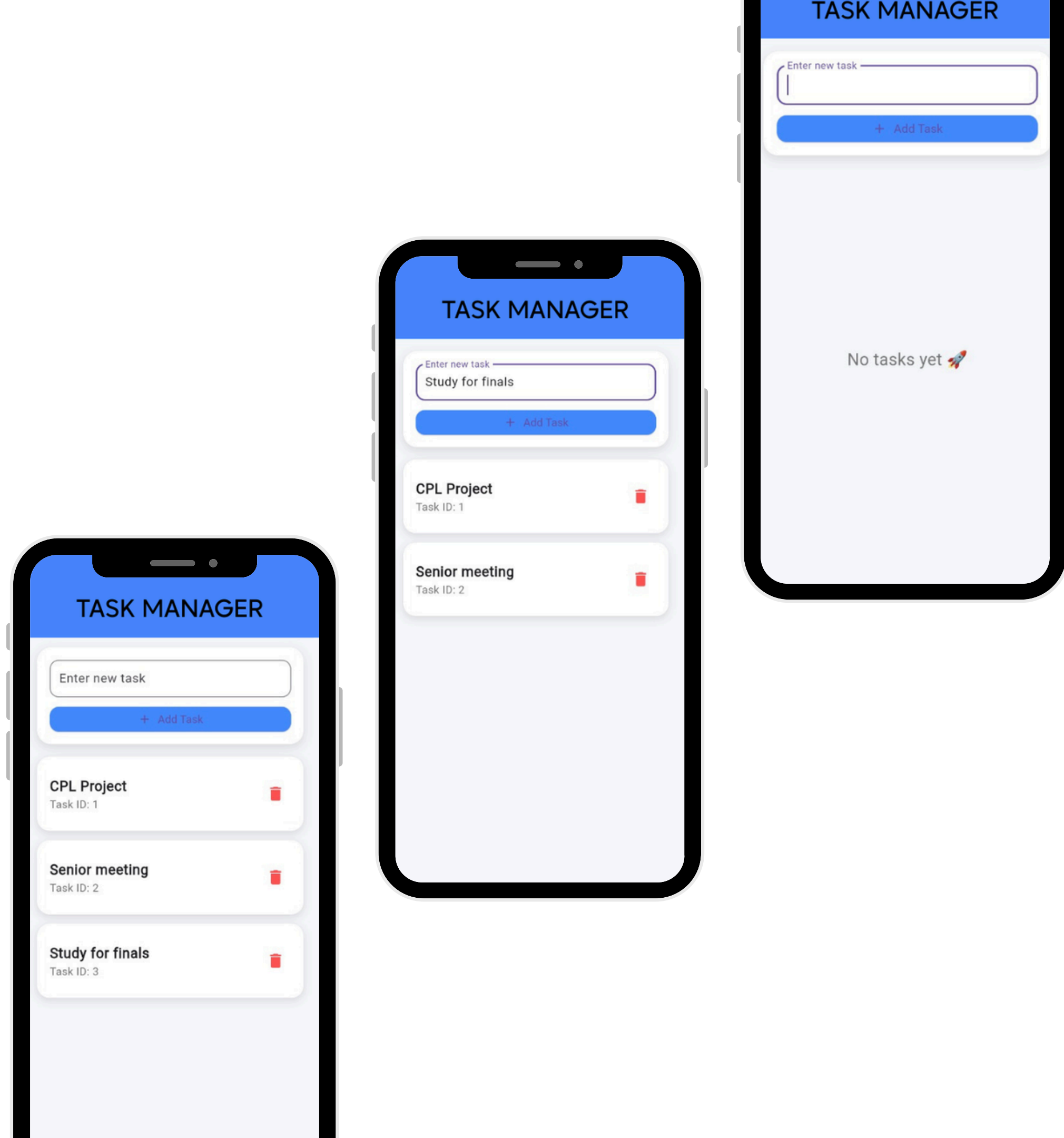


Brief Explanation of the App

Our application is a Task Manager built using Dart and Flutter.

It allows the user to:

- Add a new task by entering a title
- Display all tasks in a clean list
- Delete tasks from the list
- Automatically assign each task a unique ID

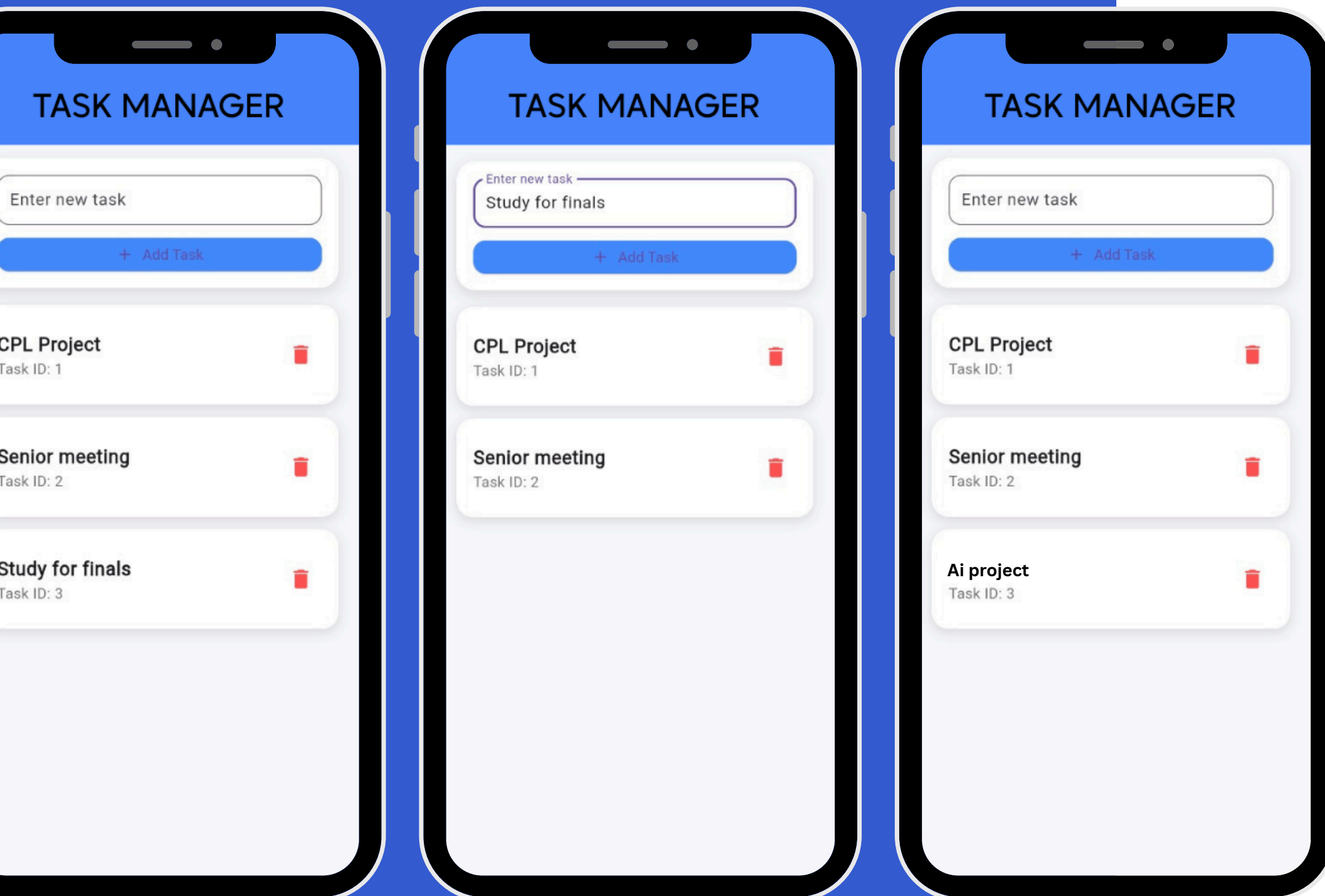
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Flutter Mobile App

This is the visual, user-friendly version where:

- Users type their task in a text field
- Press a button to add it
- See tasks displayed in a modern UI
- Press a delete button to remove any task

The Flutter version takes the same logic from the Dart code but presents it in a real mobile app interface.

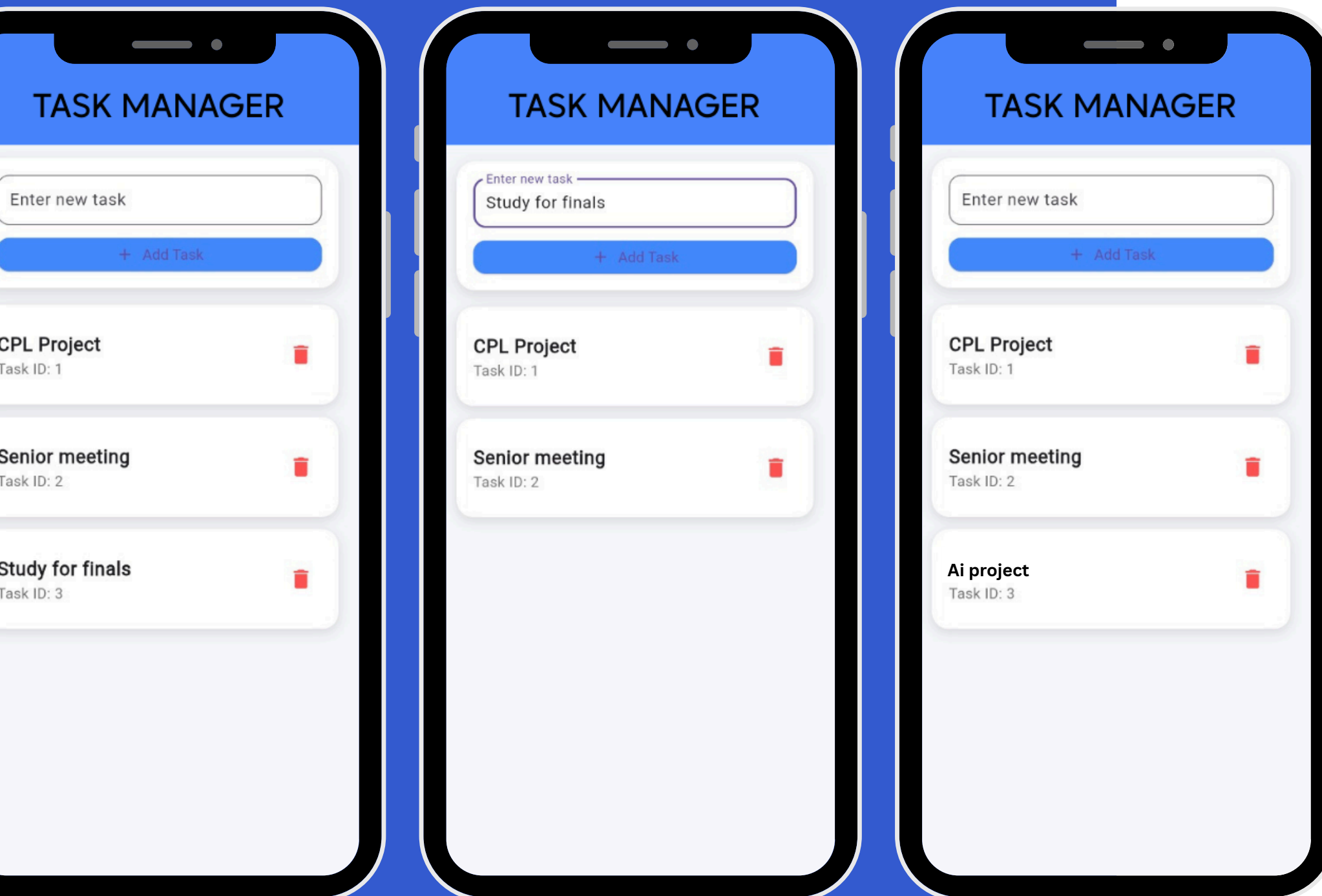


Dart vs Java

00

Readability

- Dart is cleaner, fewer brackets, less boilerplate
- Java is more verbose but very structured

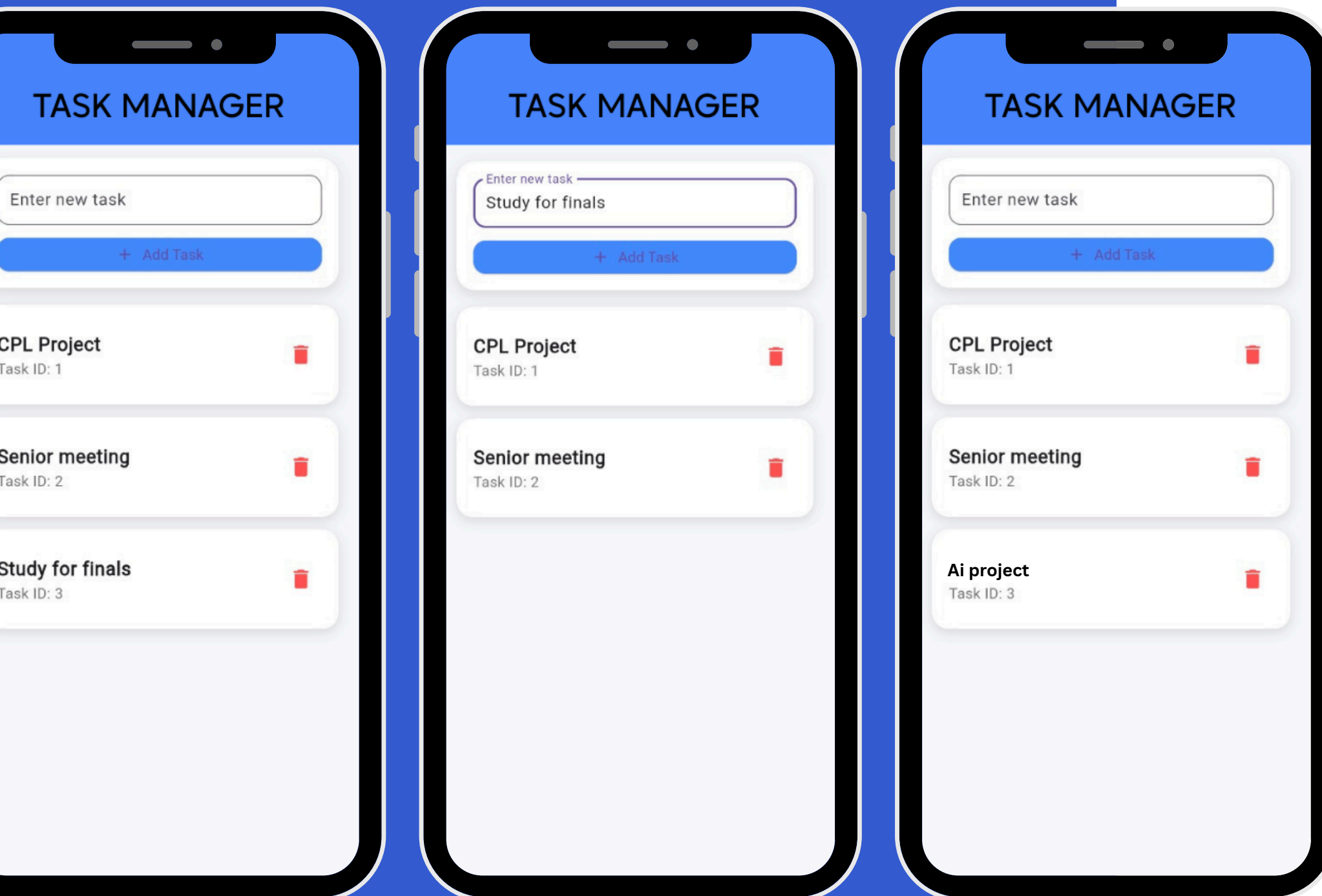


Dart vs Java

01

Writability

- Dart is quicker to write
- Supports optional types (less strict than Java)
- Java requires more setup (classes, types, etc.)

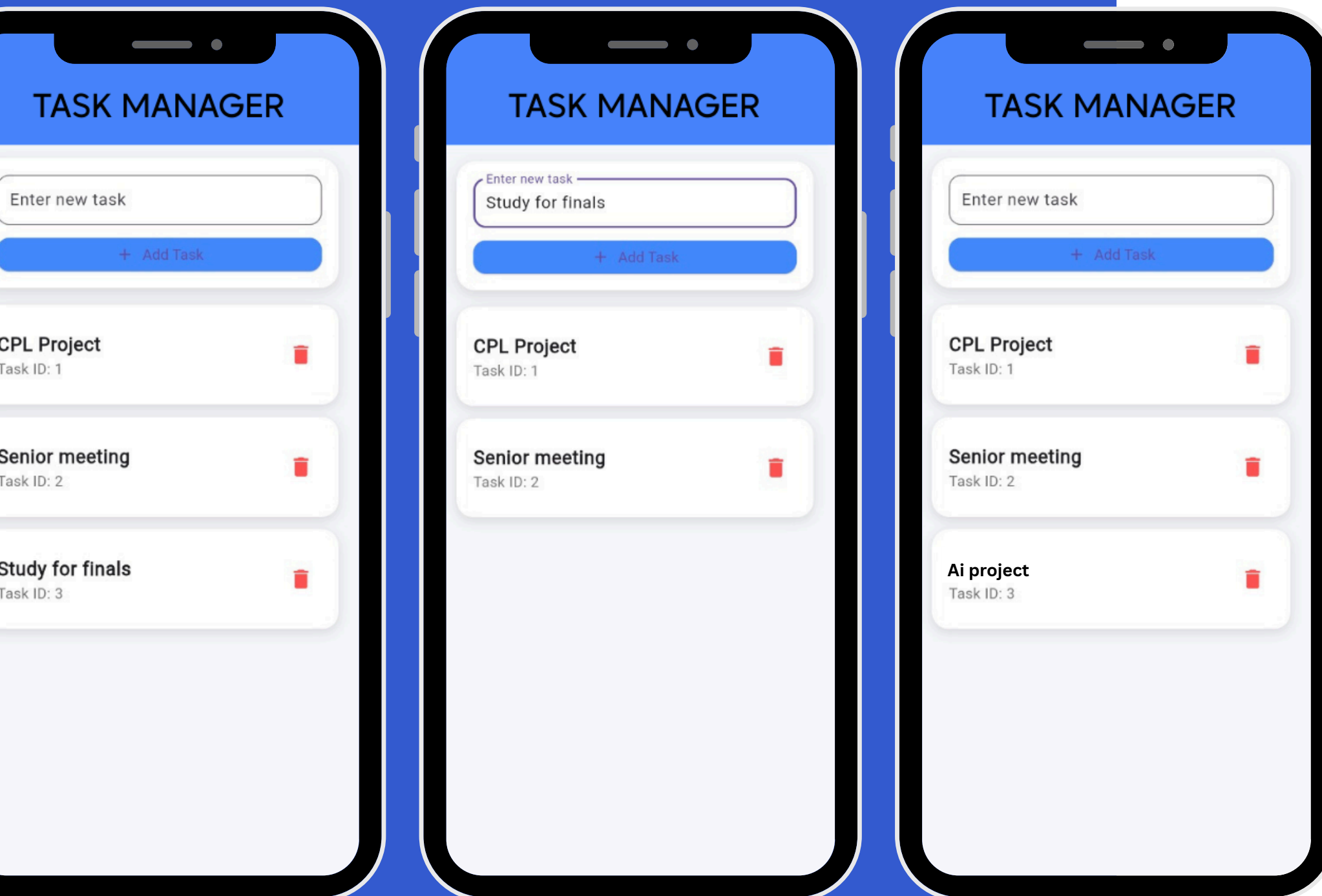


Dart vs Java

10

Reliability

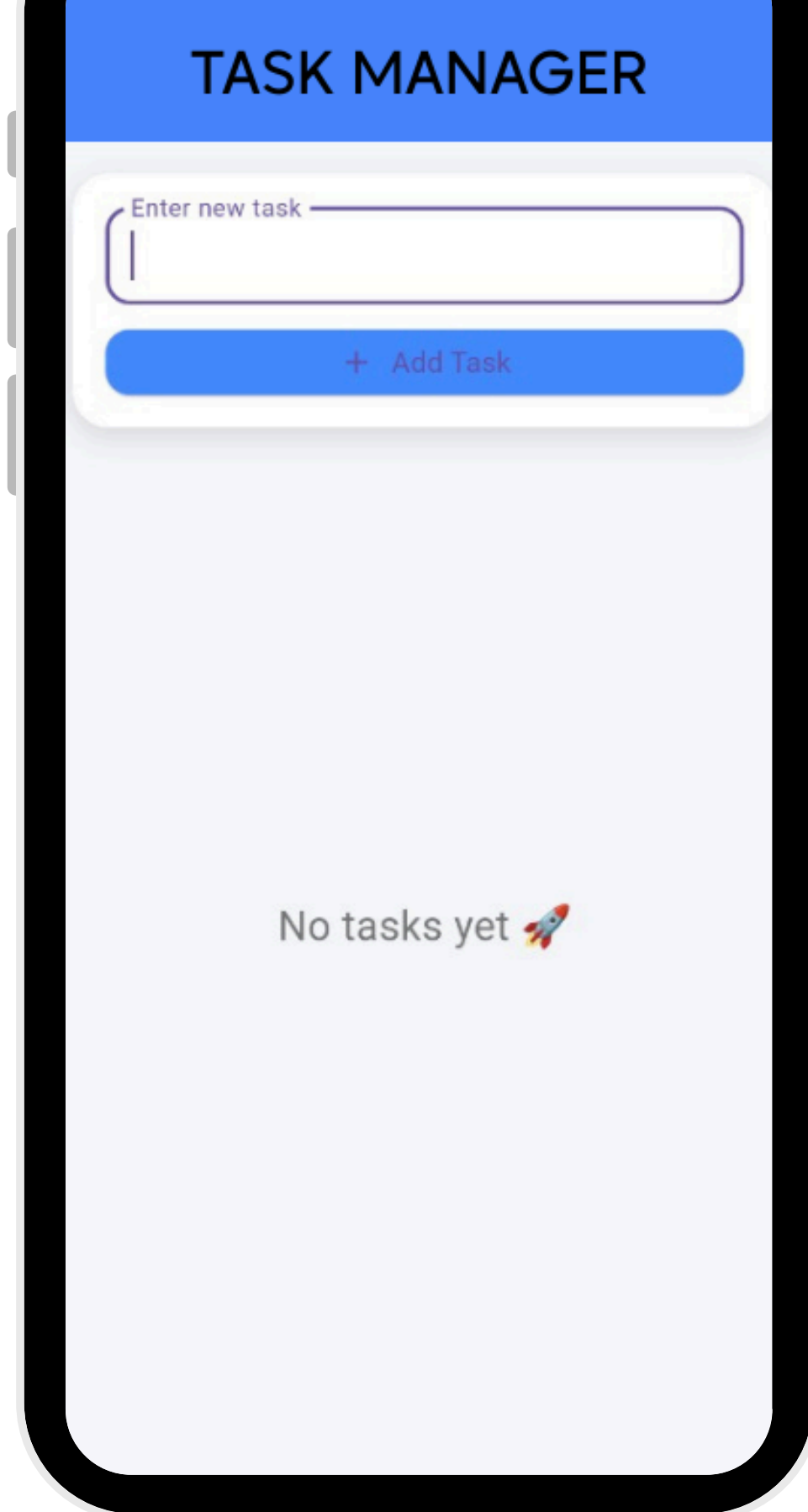
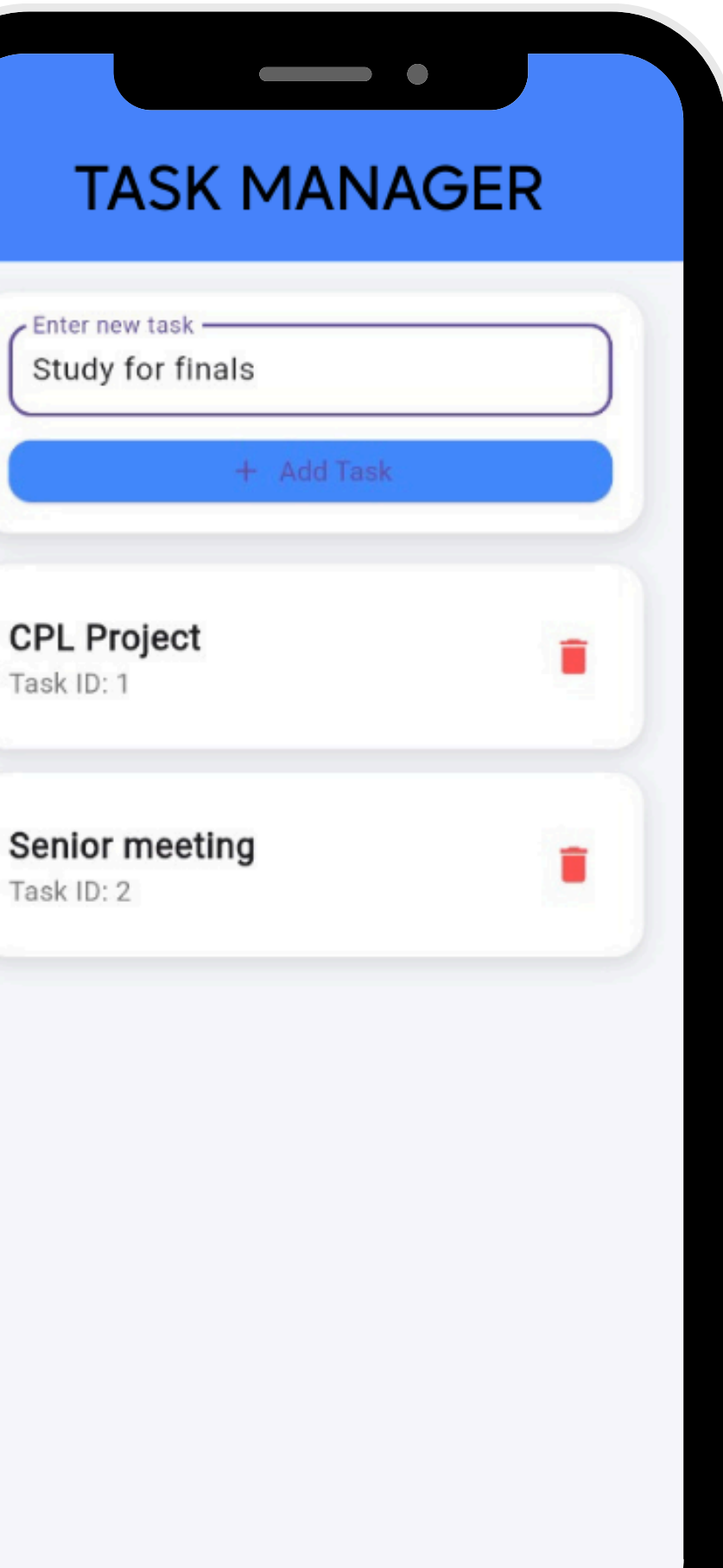
- Both have strong typing
- Java wins with strict compile-time checking
- Dart is reliable but more flexible



Dart vs Java

11 Speed

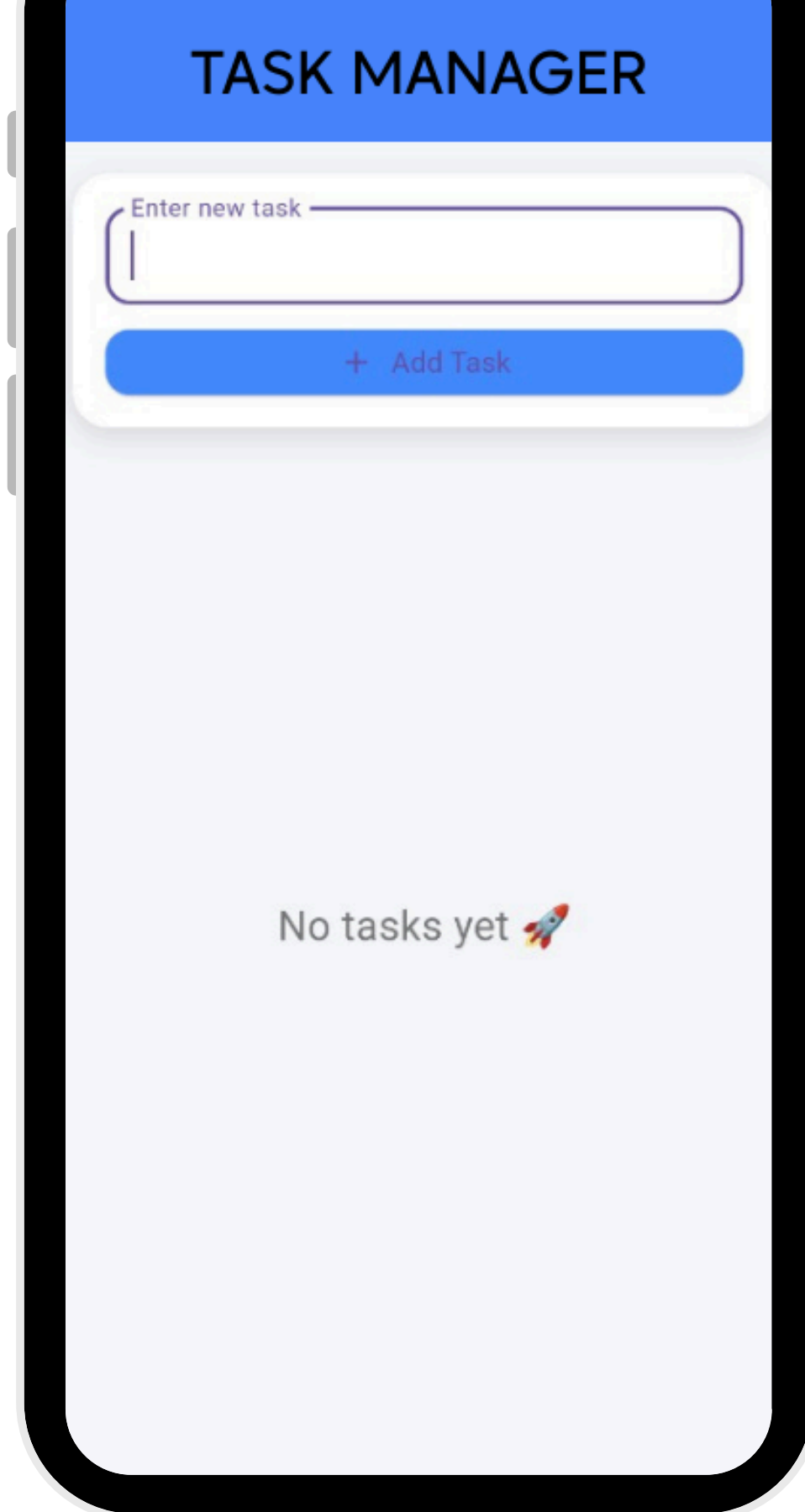
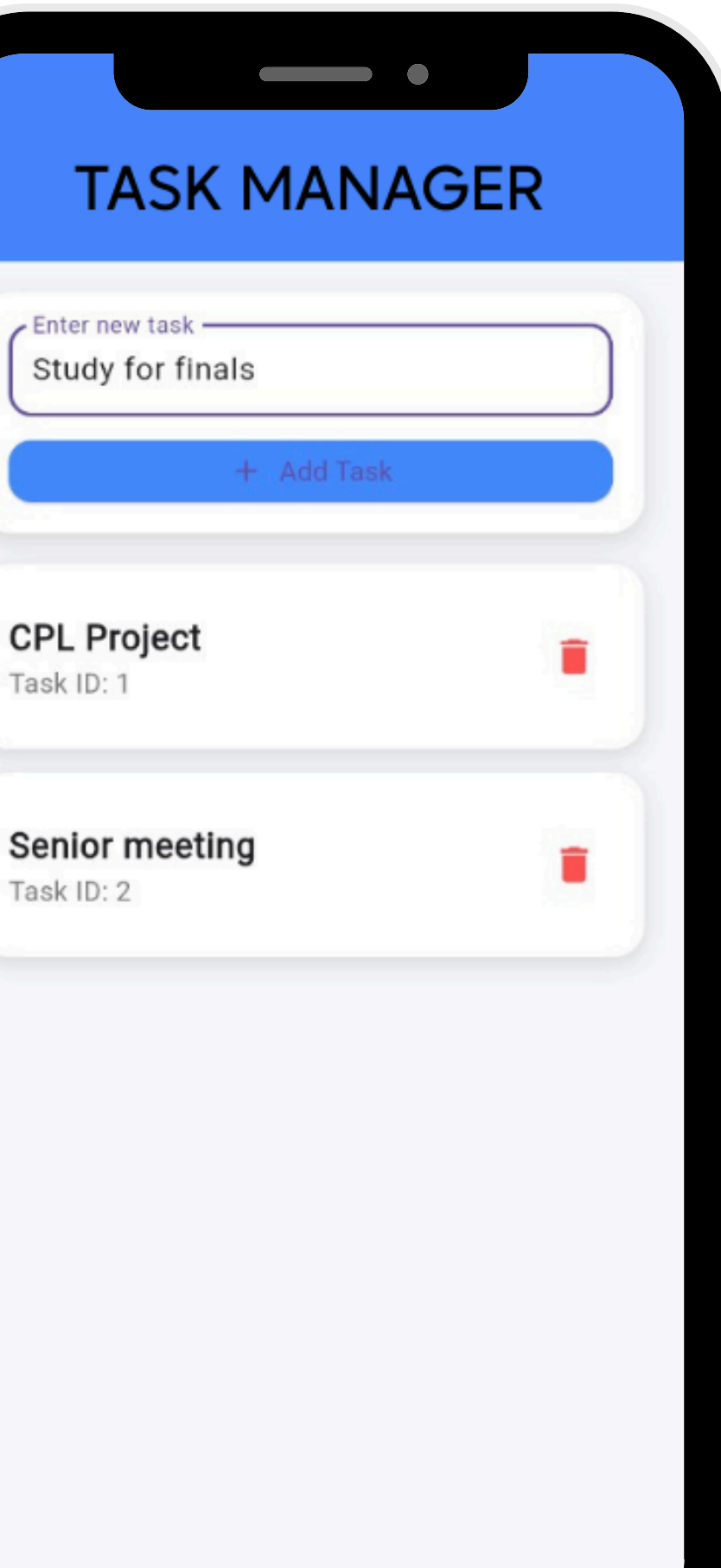
- Java is typically faster for backend systems
- Dart is optimized for UI and mobile apps (very fast with Flutter)



Challenges & Solutions

Task Class:

The class wasn't recognized at first, but we fixed it by placing small Dart files (task.dart, task_manager.dart) inside the lib/ directory, which is required in Flutter.



Challenges & Solutions

UI Design Challenges:

The first version of the UI was simple and didn't look like a real app; it needed a cleaner, more organized layout with better styling.

We improved the layout using:

- Card() for each task
- AppBar() with styled text
- spacing using Padding() and SizedBox()
- button styling with ElevatedButton.styleFrom()



Thank You

